

Project 6: Text Analysis

STAT 133, Spring 2026

About

The purpose of this assignment is twofold: 1) to perform a text analysis on a given set of text documents, and 2) to present your analysis in the form a **Quarto Website** published to Netlify.

Important Note

We are assuming that you have reviewed the learning materials on text analysis of weeks 13-14. These are based on the book *Text Mining with R: A Tidy Approach* (by Julia Silge and David Robinson):

- <https://www.tidytextmining.com/tidytext>
- <https://www.tidytextmining.com/sentiment>
- <https://www.tidytextmining.com/ngrams>

General Instructions

- Use a **Quarto Website** for your Project 6. This was covered in labs 12 and 13.
- Publish your website to **Netlify**.
 - This involves publishing the folder `_site/` of your Quarto website files.
- As usual, assume that your published website will be seen by a potential employer, boss, client/customer, admissions committee, or something like that.
- To receive full credit, please submit the following to bCourses (proj6 assignment):
 - (URL) link to your **Netlify** website.
- The full submission requires the publication of the website. In case you are not able to satisfy this requirement, you can submit a combo of `qmd` and `html` files with your text analysis in order to obtain some partial credit.

1 Text Data Sets

1.1 CSV Data Files

You will have to **choose one of the following** three data sets for your text analysis:

- Harry Potter books: `harry-potter.csv`
- Game of Thrones scripts: `game-of-thrones.csv`
- State of the Union addresses: `state-of-the-union.csv`

The corresponding CSV files are available (in their respective folder) in bcourses:

<https://bcourses.berkeley.edu/courses/1551809/files/folder/projects/project6>

1.1.1 Harry Potter

The file `harry-potter.csv` contains the text of the following 7 books written by Joanne Kathleen Rowling:

- 1) Harry Potter and the Philosopher's Stone
- 2) Harry Potter and the Chamber of Secrets
- 3) Harry Potter and the Prisoner of Azkaban
- 4) Harry Potter and the Goblet of Fire
- 5) Harry Potter and the Order of the Phoenix
- 6) Harry Potter and the Half-Blood Prince
- 7) Harry Potter and the Deathly Hallows

The CSV file has 3 columns:

- 1) **text**: text content
- 2) **book**: title of associated book
- 3) **chapter**: associated chapter number

1.1.2 Game of Thrones

The file `game-of-thrones.csv` contains the complete set of *Game of Thrones* scripts from the TV series produced by HBO.

The columns in the CSV file are:

- 1) **Date**: Original air date of the episode.
- 2) **Season**: Season number.
- 3) **Episode**: Episode number.
- 4) **Title**: Title of each episode.
- 5) **Name**: Name of character in Game of Thrones.
- 6) **Sentence**: Sentence spoken in the series.

1.1.3 State of the Union Addresses

The file `state-of-the-union.csv` involves the text of the *State of the Union Addresses and Messages* given by presidents of the U.S. from 1970 to 2021.

The dataset has four columns:

- 1) **year**: year of state of the union message
- 2) **name**: name of president
- 3) **party**: president's party
- 4) **text**: text with content of the message

1.2 Sentiment Lexicons Files

In addition to the above CSV files, there is another folder in bCourses that contains CSV files of the **sentiment lexicons** from the "textdata" package:

- Bing's lexicon: `bing.csv`
- Afinn's lexicon: `afinn.csv`
- Loughran's lexicon: `loughran.csv`
- NRC lexicon: `nrc.csv`

! Do not use `get_sentiments()`

Don't use the function `get_sentiments()` to obtain a sentiment lexicon since this is an interactive function that requires input from the user. Instead, use one or more of the lexicons provided in CSV format.

2 Text Analysis

This assignment involves conducting the two kinds of text analysis discussed in class, namely: 1) Frequency Analysis, and 2) Sentiment Analysis. Optionally, you can perform some of the other analysis described in the book [Text Mining with R: A Tidy Approach](#), for instance TF-IDF (chapter 3), or correlation analysis (chapter 4).

2.1 Frequency Analysis

Recall that this is most basic type of text analysis, also referred to as "bag-of-words analysis". The main idea is to identify the most frequent words or tokens (1-grams, 2-grams, etc). This analysis is usually done after removing stop words.

You can compute different kind of frequencies, for example but not limited to:

- most common words in a given book or episode or address,
- most common words per book or per episode or per address,
- most common words associated to a certain character,
- trends of most common words over a period of time (years) or over the chapters of a book, etc.

2.2 Sentiment Analysis

The goal here is to show how tone shifts or differs across the text.

- **Dictionary Comparison:** Perform sentiment analysis using at least two different sentiment lexicons (e.g., `bing` for binary and `nrc` for emotions) and comment on why the results differ.
- **Sentiment Trajectory:** Divide the text into “chunks” (e.g., every 80 lines or by chapter or by episode or other meaningful category) and plot how the net sentiment changes from beginning to end.
- **Contribution Analysis:** Identify which specific words contributed the most to the sentiment score (e.g., a bar chart showing the words with the highest (sentiment * count) impact).

3 Website Specifications

The focus of the website should be on **communication**.

- **Multi-page Structure:** Use at least three pages:
 - **Home page** describing the purpose and scope of your analyses, as well as a description of the chosen data set.
 - **Frequency Analysis page** containing the analysis (narrative, code, outputs) of the frequency analysis.
 - **Sentiment Analysis page** containing the analysis (narrative, code, outputs) of the sentiment analysis.
- **Code Visibility:** Use `code-fold: true` in your code chunks so the focus remains on the prose and plots, but the R code is accessible to the interested reader.
- **Data Visualizations:** Each of the analysis pages should contain 4-6 high-quality graphics:
 - Ideally, all **graphics should be interactive** (e.g. plotly, or ggiraph) unless a given plot can only be done in static format.
 - Use of color wisely and purposely, not just as *eye candy*.

The content should be well organized and presented. This means that **any** STAT 133 member (students and staff) should be able to visit your website and follow the *journey* of your analysis without having to wonder what a certain section is used for, or what kind of analysis has been done, or what a given output means.